Claude Code Black Belt

Zugzwang: **who decides when the agent has to move?**

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Stay to the end. There is a gift: **a starter repo to take home.**

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How often do you work with a coding agent?

Poll 1 of 4

1	2	3	4
almost never	rarely	weekly	daily

Claude Code or any other. Type the number in the chat.

What do you work with?

Poll 2 of 4

1	2	3	4	5
Claude Code	GitHub Copilot	Codex	Cursor	something else

Several? Type all numbers, for example: 1 3

What have you set up?

Poll 3 of 4

1	2	3	4	5
chat only	CLAUDE.md or AGENTS.md	skills	hooks	subagents or agent teams

Several? Type all numbers, for example: 2 4

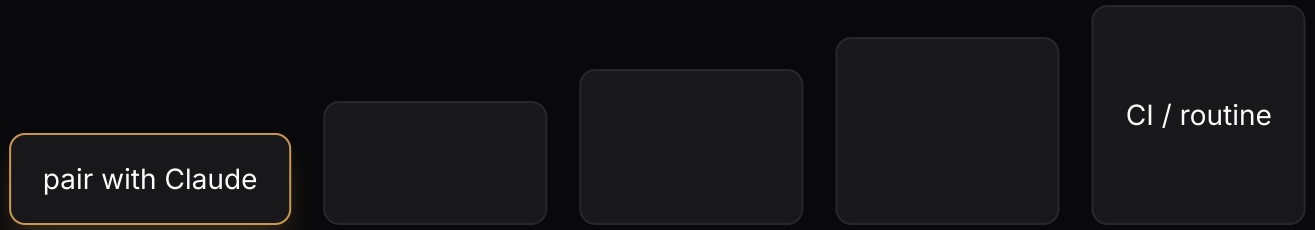
Has your agent ever changed an existing test?

Poll 4 of 4

1	2	3	4
yes, I noticed	yes, noticed later	no	don't know

Type the number in the chat. An honest answer helps.

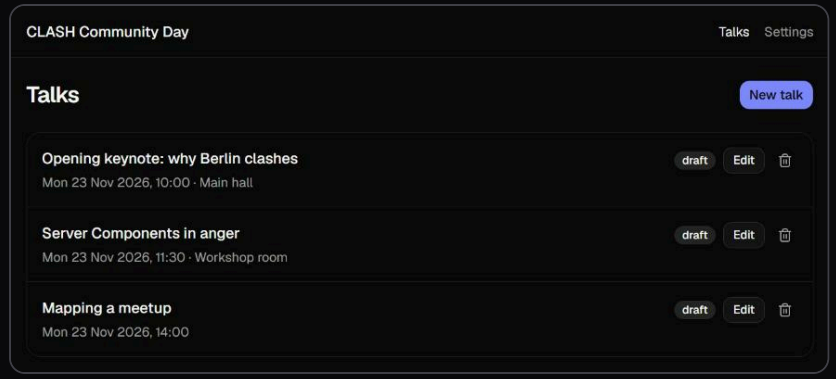
Where do you stand today?



Today's live demo starts here.

How much does the agent do without you?

The app: one question, two answers



clash-conference: the program of a conference. A talk has a title, a start and a room.

```
Can a talk start in the past?
Can two talks share a room and a start?
Name the file behind each answer.
```

Both: yes. Nothing checks either.

Fluent is not the same as true

```
Can Claude still change a test with sed -i when Edit(tests/**) is denied?
```

guessing · "quick question"

"Yes: the deny only covers the file tools, not Bash."

No tool call. From training. Sure of itself. Wrong.

grounded · "check the docs first"

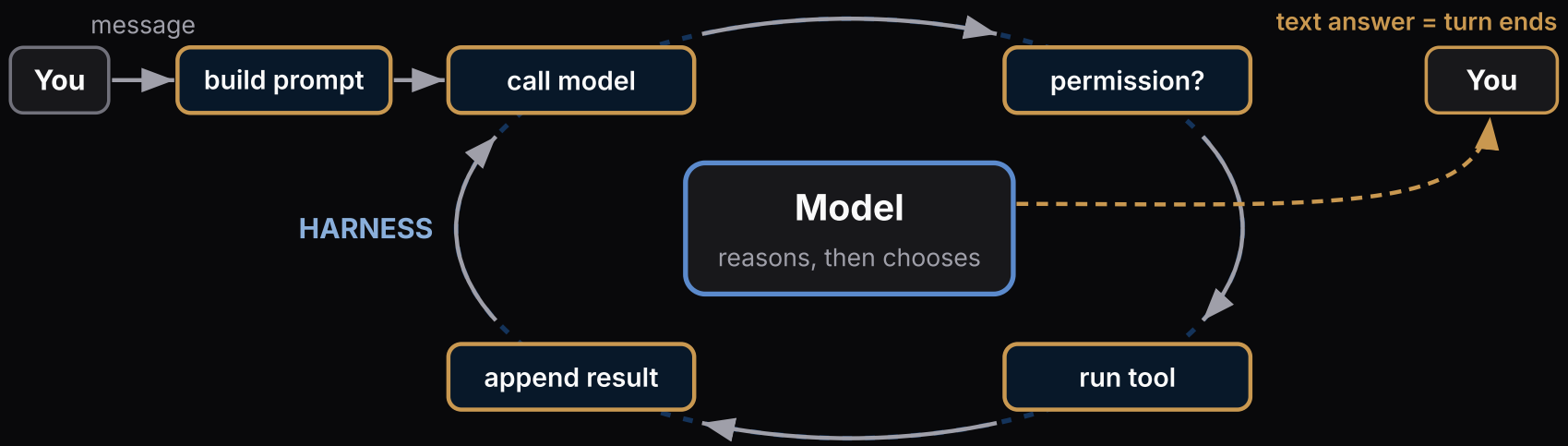
"No: Edit deny rules also cover sed, tee and > in Bash."

Read the permissions docs, then answered. Quotes the source.

The model always produces something that reads well.

It reads your code on its own. It guesses where it thinks it already knows.

The loop

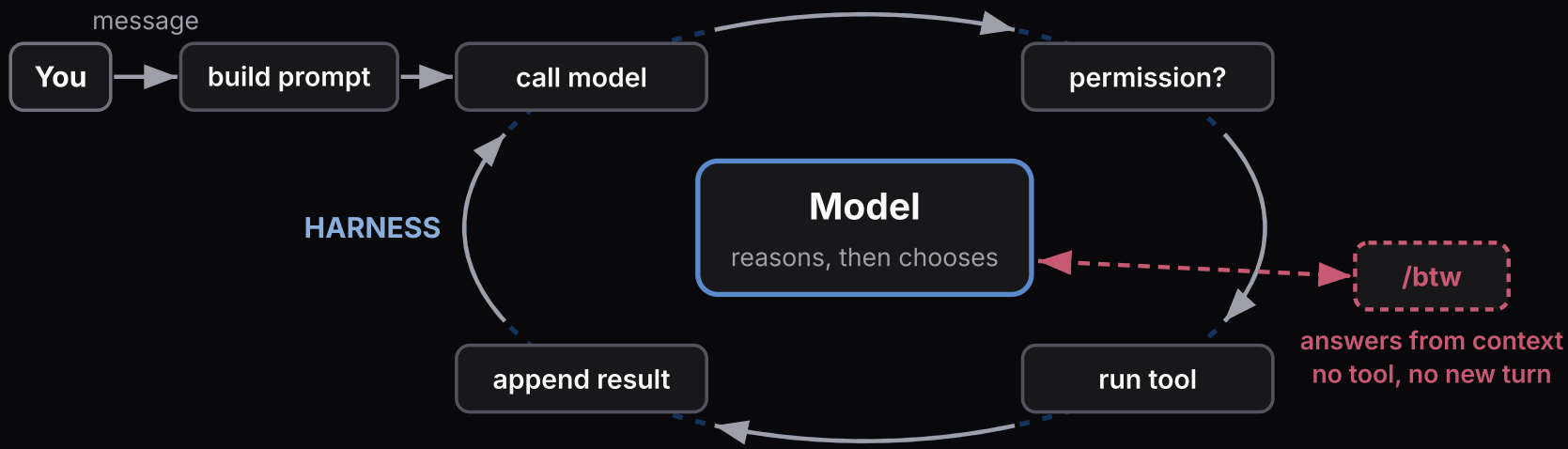


You write. The harness builds the prompt. The model chooses a tool or answers.

The harness runs the tool and feeds the result back. Until the model answers.

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A question that skips the loop



Answered from what is already in the window. No tool call, no new turn.

Thinking time: does more effort make a better move?

`--effort low`

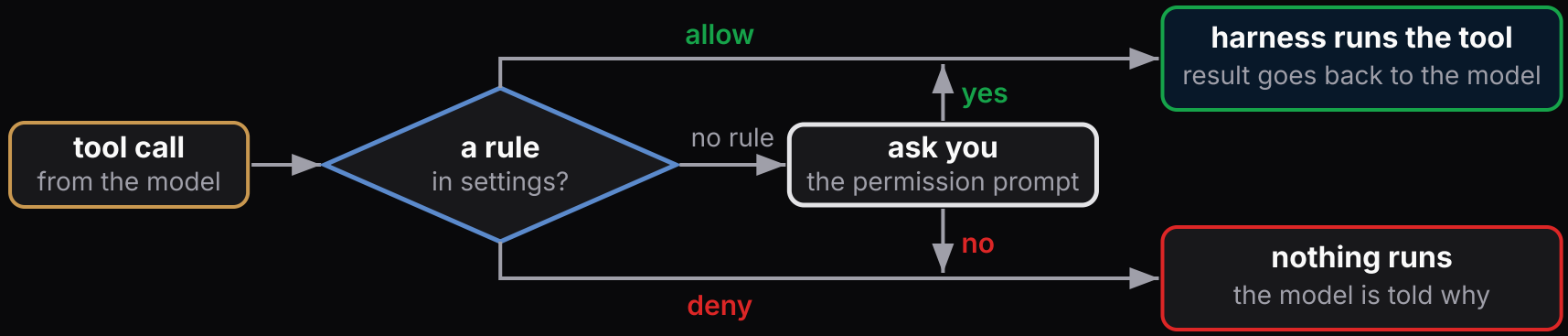
`--effort high`

Same file. Same prompt. Same model. Four hidden bugs.

The race starts now. The result comes later.

Effort sets how much the model thinks on every step.

Permissions



MODE · how often the gate asks you

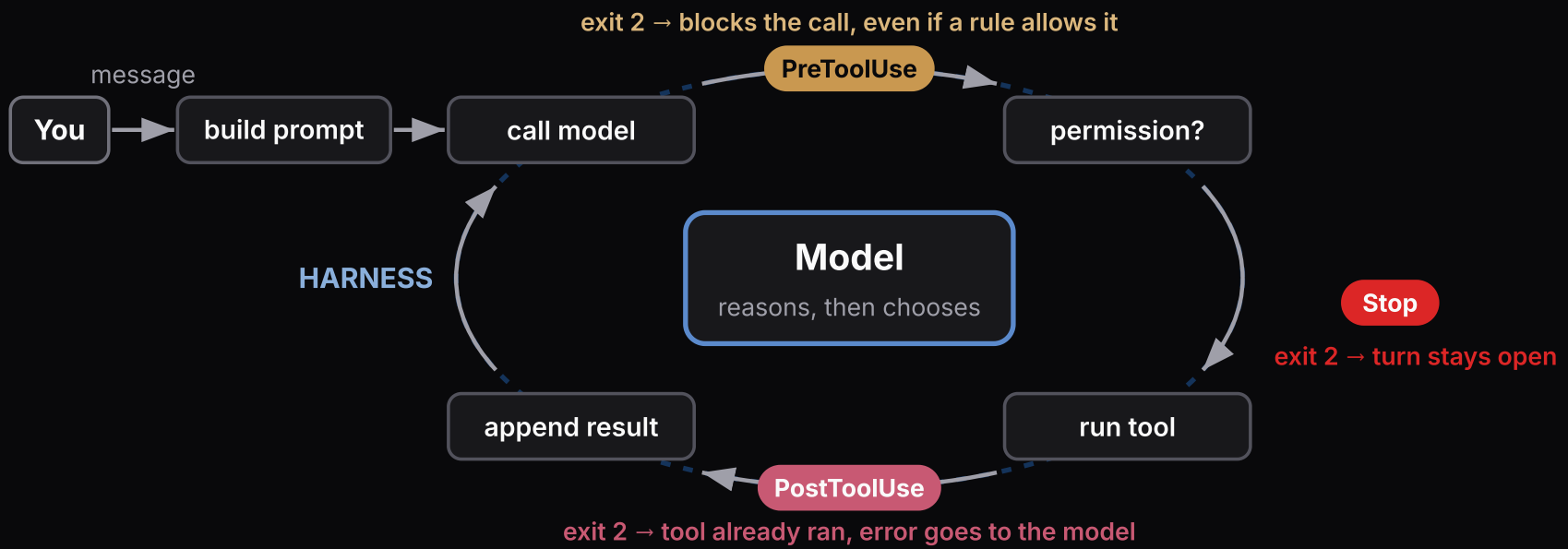
manual | accept edits | plan | auto | bypass

bypass: rarely asks — containers and VMs only

The model asks. The harness decides: a rule from settings, or you.

The mode sets how often the gate asks you.

Hooks



Your own shell command on the ring: before a tool, after it, or at the exit

Before a tool, exit 2 blocks it. After a tool, exit 2 gives the model the error.

Zugzwang: update the test, or fix the logic?

Careless	Engineered
“update the test so the suite passes”	`“change lib/validation.ts only, not the tests”`
Claude edits the assertion	Claude edits the logic
green, rule broken	green, rule holds

Same failing test. Which file was allowed to change?

Advice vs. law

ADVICE

CLAUDE.md: Never edit a file in `tests/` to make a failing test pass.

Can be forgotten.

LAW

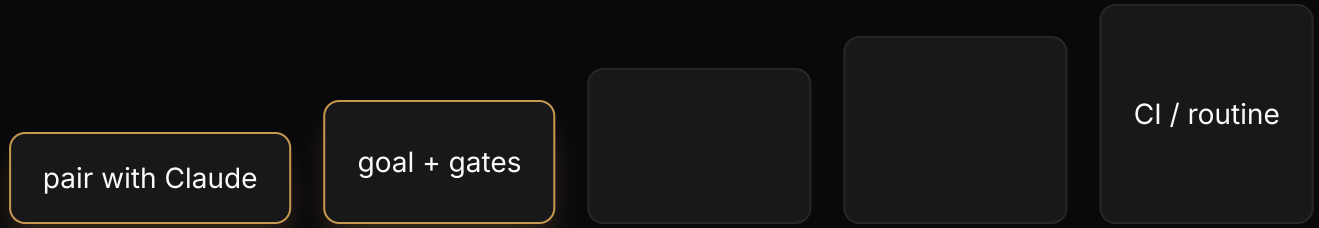
Hook `guard-tests`: a new test passes. An old test: Claude explains, you approve.

Hook `guard-db`: blocks `db:seed`, `db:reset`, `setup` and `rm dev.db`.

Runs outside the model, every time.

```
git status --short -- tests/  ·  ?? = new  ·  M = look again
```

Where do we stand now?



Two steps climbed today: we paired with Claude, then hook and test set the gates.

The race: low vs. high

Today: five runs per level

	correct	false +	median output tokens
`low`	…	…	…
`high`	…	…	…

More thinking is a dial, not a guard. Safety came from the hook and the test.

Live result: in the terminal.

Subagents: three checkers, one prompt

```
Use three Explore subagents in parallel, read only:
1. Does the app refuse a start in the past?
2. Does it refuse two talks in one room at one start?
3. Which files under tests/ changed?
   (git status --short -- tests/)
Each answers in one line with file:line.
```

`Explore 1` · start in the past? · read only

`Explore 2` · one room, one start? · read only

`Explore 3` · tests/ changed? · read only

Back: three lines with file:line. What they read stays in their windows.

Each subagent works in its own context window. Only its answer comes back.

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Isolated agents: Claude in a box

```
// the core of .devcontainer/devcontainer.json
{
  "image": "mcr.microsoft.com/devcontainers/javascript-node:22",
  "features": {
    "ghcr.io/anthropics/devcontainer-features/claude-code:1.0": {}
  }
}
```

1 · `claude setup-token` → `.devcontainer/.env` (git-ignored)

2 · VS Code: `Dev Containers: Reopen in Container`

3 · terminal in the container: `claude`

Commands run in the container. Your project files stay on your machine.

Trusted repos only: a malicious project can take what is in the box, even ~/.claude.

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What we added to the box

token hidden

Commands Claude runs don't see it. Claude Code keeps it for its own calls.

```
CLAUDE_CODE_SUBPROCESS_
ENV_SCRUB=1
```

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no reading secrets

Deny rules in `.claude/settings.json`.

```
.env* ·
~/.claude/.credentials.json ·
~/.ssh · ~/.aws
```

docs

sandbox on

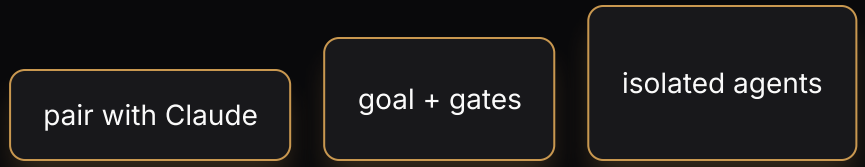
Windows has no Bash sandbox. The Linux box does.

```
"sandbox": { "enabled": true }
```

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Three layers. Test them in your box: a script that opens .env must fail.

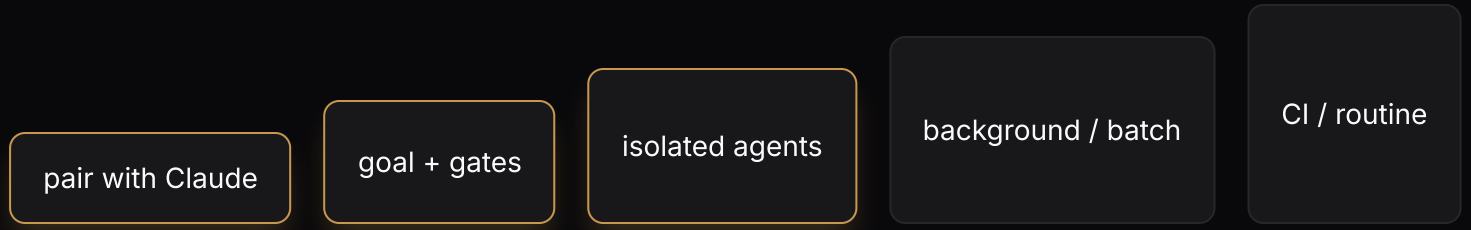
Autonomy is earned by verification



More freedom needs stronger evidence, isolation and limits.

Do not scale autonomy beyond your ability to detect and contain errors.

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The steps where nobody watches

`/batch`

Many separate units, each ending as a tested pull request.

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CI

A pull request opens. A separate agent audits it. Nobody watches.

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Mastering Claude Code

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nextacademy.io/de/workshops/claude-code

You stayed to the end

Your gift: **a repo with the guard-tests hook, a verification kit, a course task and a cheat sheet.**

`github.com/nextacademy-io/claude-code-webinar`



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